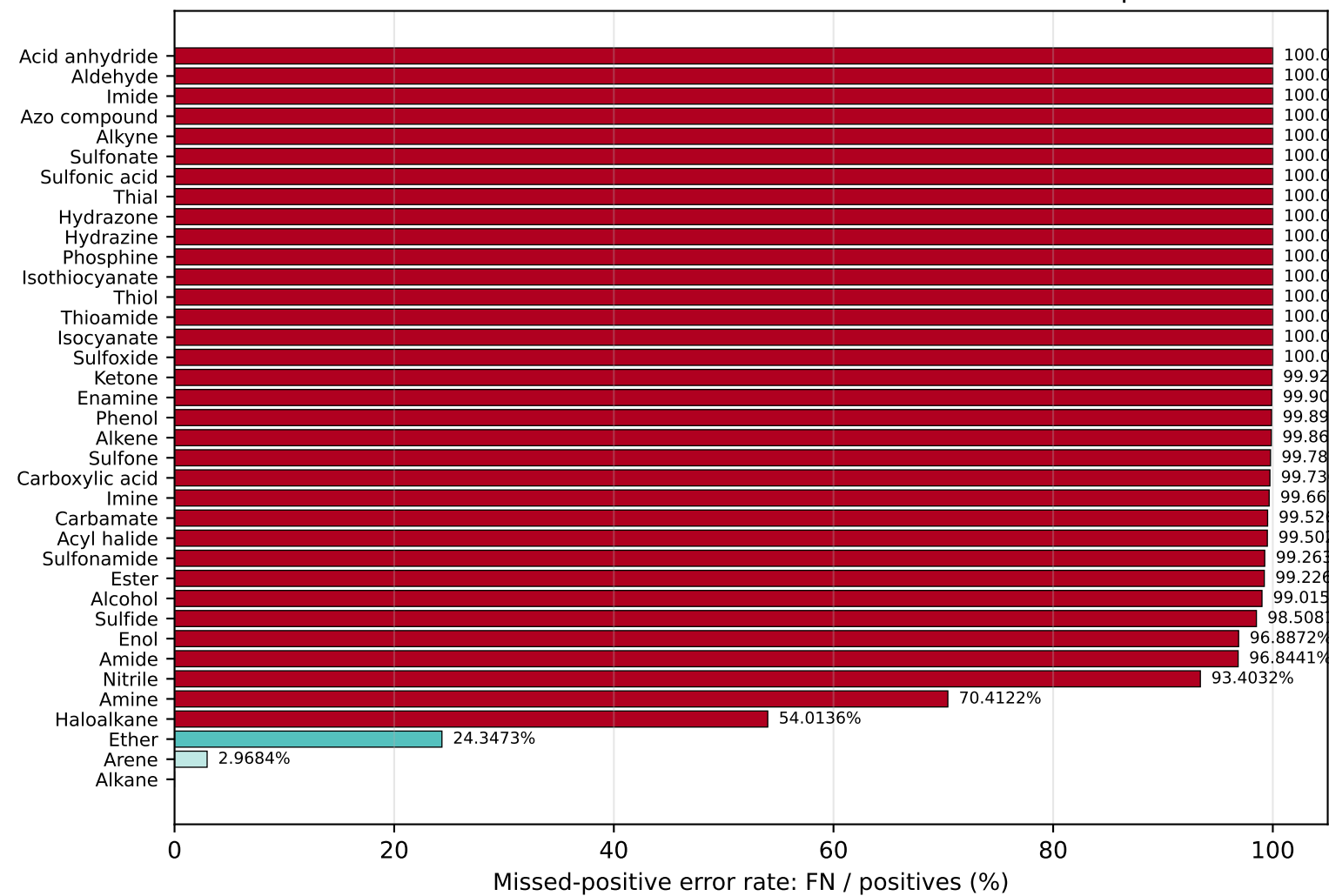
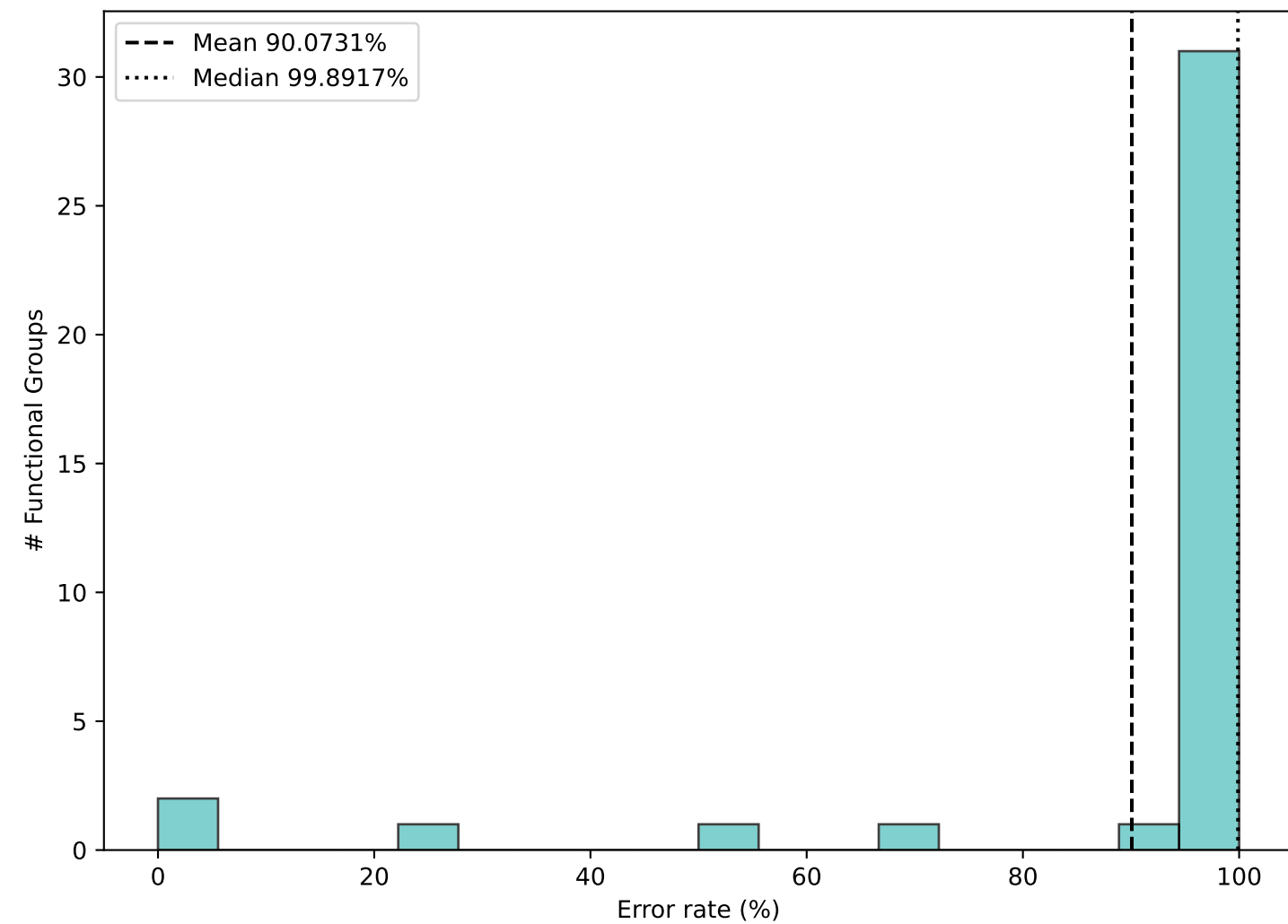


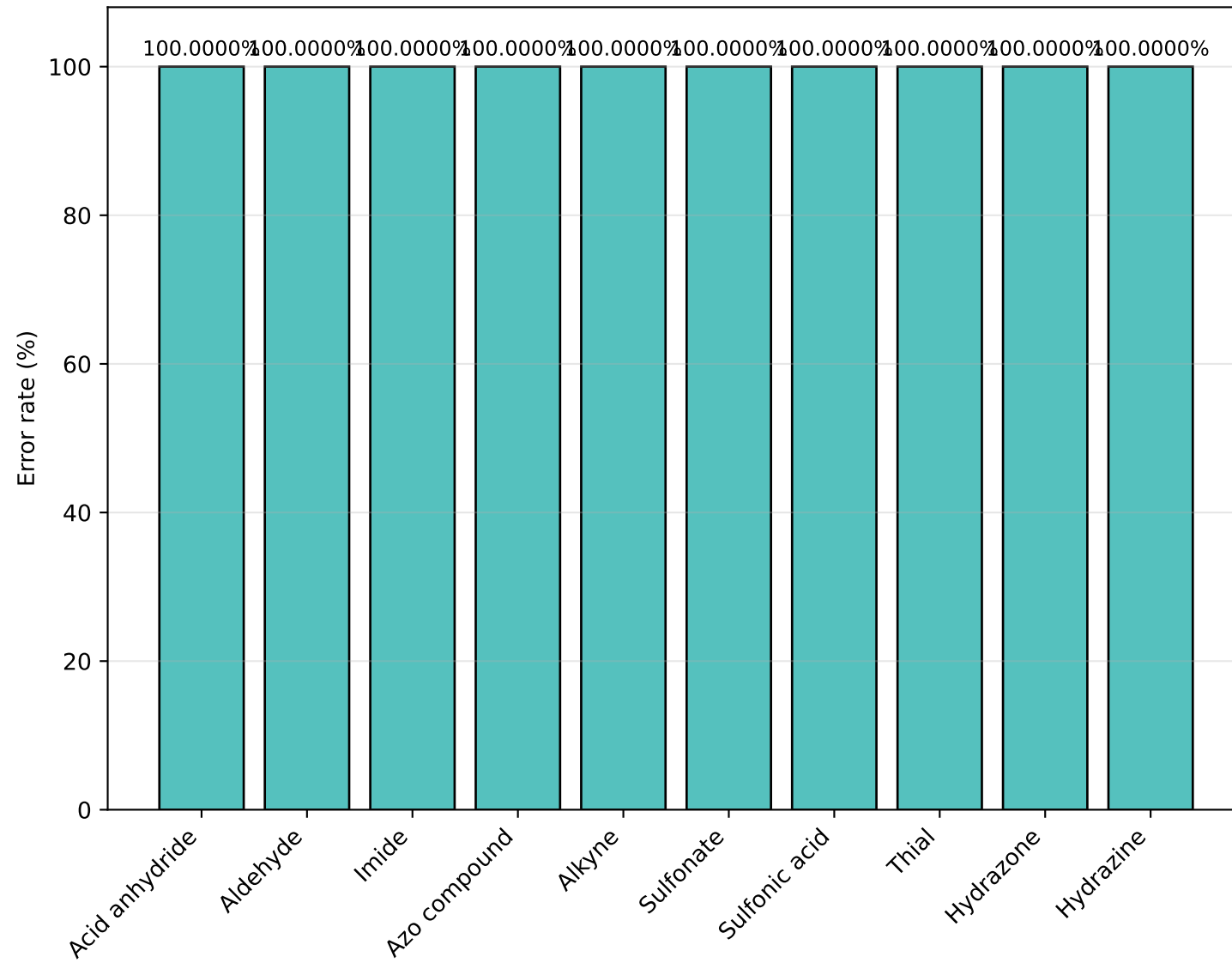
XGBoost MSMS+ Validation Set - All 37 Functional Groups



Error-Rate Distribution



Top 10 Problem Groups



XGBOOST MSMS+ VALIDATION SET ERROR ANALYSIS

Global metrics:

F1 micro:	0.6350
F1 macro:	0.1055
Precision micro:	0.8067
Recall micro:	0.5235
Hamming accuracy:	0.9223

Error-rate statistics:

Mean:	90.0731%
Median:	99.8917%
Std:	26.1145%
Max:	100.0000% (Acid anhydride)
Min:	0.0000% (Alkane)

Problem classes:

Severe (>50%):	34 groups
High (20-50%):	1 groups
Moderate (5-20%):	0 groups
Low (<=5%):	2 groups